

Impacts of forecast based anticipatory action: Evidence from flooding in Nepal and Bangladesh

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April 19, 2026



Forecast-based anticipatory action

- Key question for humanitarian response: What is the impact of delivering cash at the moment of a climate-related extreme weather disaster?
- Impacts of cash earlier could be bigger or smaller than cash later because:
 - Cash earlier can help to avoid worst outcomes when marginal utility of consumption is highest (eg supporting food security when food stocks are damaged, stress when people are coping with disasters)
 - Cash earlier can facilitate recovery (financing next season's planting without high cost loans)
 - However, cash right after disaster hits could be less effective because of missing markets or harder to target because we don't know who is affected yet immediately following a disaster

- Two impact evaluations in Nepal and Bangladesh test: **Can anticipatory cash transfers delivered as quickly as possible after a flood forecast is triggered help households cope with the impacts of the shock (when compared to later transfers)?**
- We use two RCTs to measure impacts of cash transfers earlier vs later on food security, stress, and coping strategies during three periods after the flood event

- Supports results from Adeyanju et al (2023), Gros et al (2020), Lane (2023), Pople et al (2021), Pople et al (2023), and others that pre-positioning support ahead of disasters can help with recovery, but room for further adjustments to facilitate subsequent livelihoods related investments.
- Also relates to broader literatures on impacts of cash transfers (Kondylis and Loeser, 2021)
- We use randomized assignment of timing to study whether, and for what outcomes, additional benefits accrue specifically from acting early in a context of a rapid onset event.

How does FbAA work?– eligibility and registration

WFP conducts a for-purpose targeting and registration at three levels:

- **Regional:**

- River basins at risk of flood and high vulnerability

- **Municipal**

- Coverage under Global Flood Awareness System (GloFAS)-based trigger mechanisms

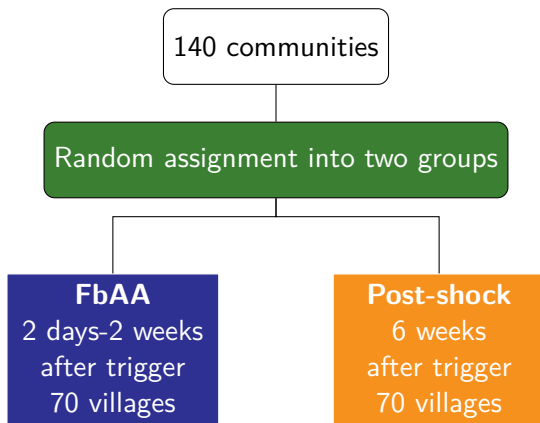
- **Household-level**

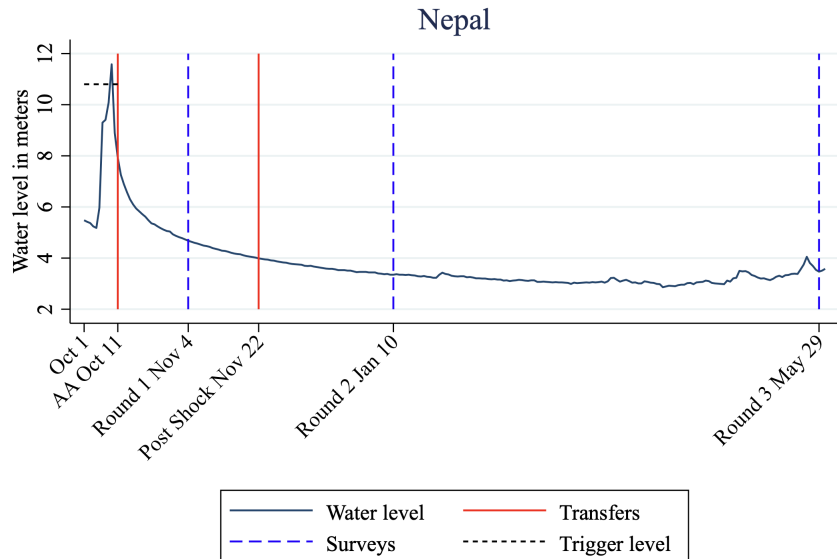
- Coordinated with local governments. Vulnerability factors that are considered include position relative to rivers, engagement in agriculture, demographics associated with vulnerability

How does FbAA work?—implementation

- Nepal (12,500 households)
 - One time distribution of \$111
 - Physical delivery of cash
- Bangladesh (90,000 households)
 - One time distribution of \$43
 - Mobile money

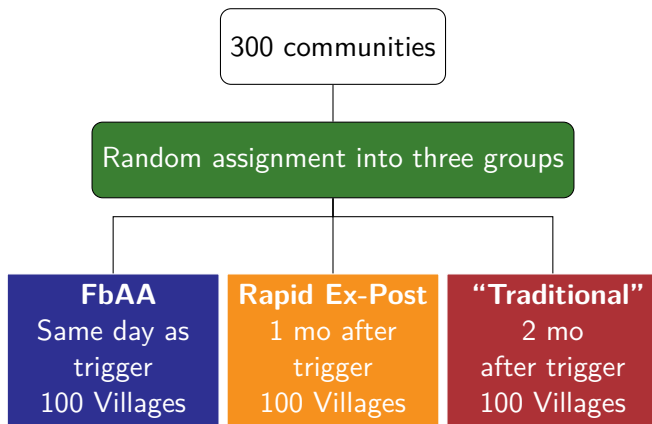
Nepal Design

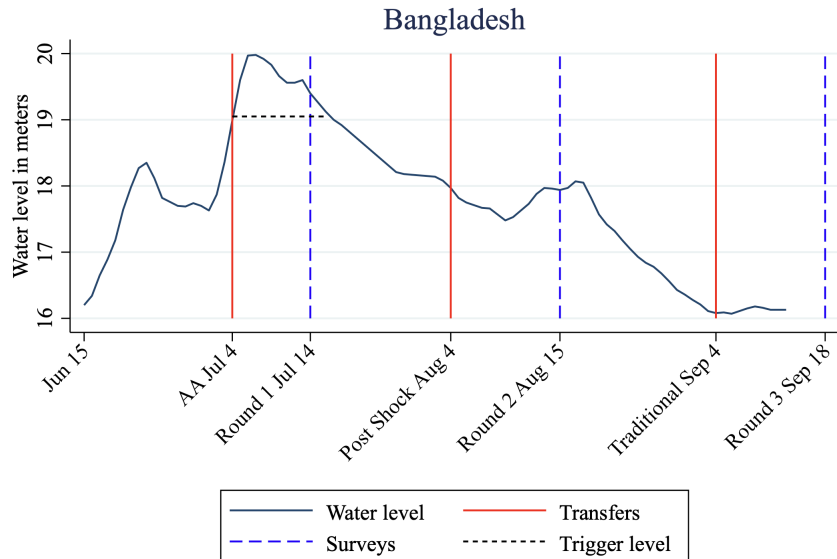




Chisapani station measured water level.

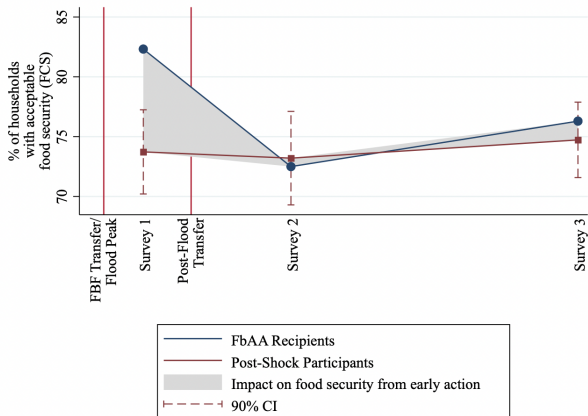
Bangladesh Design





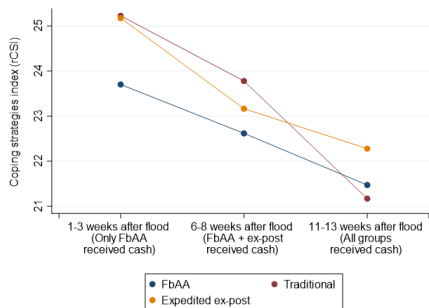
Bahurabad station measured water level.

Nepal-Acceptable food security



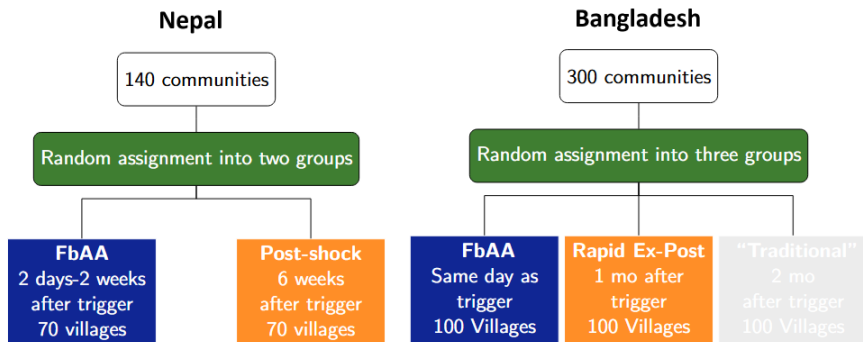
Impact of AA on food security mostly comes from acting quickly. Better food security in the early acute stress period when only the AA group has received transfers.

Bangladesh-Reduced Coping Strategies Index



Similarly, in Bangladesh, we find improvements in less negative coping strategies mostly from shifting impacts forward to the flood period.

Pooling FbAA vs earliest Ex-post



In both countries, this is gives us:

- R1: Cash vs nothing right after the flood peak
- R2: "older" FbAA cash vs recent Ex-post cash as flood recedes
- R3: FbAA vs ex-post later on

Impact of early transfers from two countries

For Nepal:

$$y_{ijkt} = \beta_t FbAA_j * Round_t + \theta_{tk} + \epsilon_{ijkt} \quad (1)$$

For Bangladesh:

$$y_{ijkt} = \beta_t FbAA_j * Round_t + \delta_t Trad_j * Round_t + \theta_{tk} + \epsilon_{ijkt} \quad (2)$$

To compare FbAA against the earliest ex-post transfer, we'll show β_t for each round and then pool by inverse variance weights across countries, ignoring δ_t .

	Round 1		Round 2		Round 3	
	FCS	rCSI	FCS	rCSI	FCS	rCSI
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Nepal						
Anticipatory Action	2.883**	-0.959*	-0.063	0.220	0.386	
	(1.200)	(0.557)	(1.347)	(0.339)	(1.077)	
	[0.018]	[0.087]	[0.963]	[0.518]	[0.720]	
	N=2212	N=2212	N=2365	N=2365	N=2408	
Panel B: Bangladesh						
Anticipatory Action	0.542	-1.337**	1.168	-0.397	0.455	-0.887**
	(0.707)	(0.562)	(0.751)	(0.566)	(0.547)	(0.429)
	[0.445]	[0.018]	[0.121]	[0.484]	[0.405]	[0.039]
	N=4721	N=4721	N=4761	N=4761	N=4485	N=4485
Panel C: Inverse-Variance Weighted-Average						
Anticipatory Action	1.146*	-1.146***	0.876	0.057	0.441	
	(0.609)	(0.396)	(0.656)	(0.291)	(0.487)	
	[0.060]	[0.004]	[0.182]	[0.846]	[0.365]	
Post-flood mean:NEP	47.9	6.5	47.9	2.6	47.4	
Post-flood mean:BGD	47.0	24.2	48.0	23.4	50.9	21.7
Post-flood mean:Pooled	47.4	16.8	48.0	14.6	49.3	
Strata X Round FE	Yes	Yes	Yes	Yes	Yes	Yes
N Clusters	428	428	438	438	438	438
N Observations	6933	6933	7126	7126	6893	4485

Notes: This table presents the impact of being in the FbAA group relative to the post-flood group on Food Consumption Score (FCS) and Reduced Consumption Strategies Index (rCSI), across the three survey rounds. Panel A shows results for Nepal only. Panel B shows results for Bangladesh only. Panel C shows an inverse-variance weighted average of the results from Nepal and Bangladesh. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels.

	Round 1		Round 2		Round 3	
	PHQ4 (1)	Cantril Ladder (2)	PHQ4 (3)	Cantril Ladder (4)	PHQ4 (5)	Cantril Ladder (6)
Panel A: Nepal						
Anticipatory Action	-0.097 (0.070) [0.167] N=2212	0.154* (0.084) [0.068] N=2212	-0.059 (0.069) [0.395] N=2365	0.084 (0.086) [0.330] N=2365	0.022 (0.061) [0.716] N=2408	0.009 (0.067) [0.890] N=2408
Panel B: Bangladesh						
Anticipatory Action	-0.104* (0.057) [0.068] N=4721	0.195*** (0.066) [0.003] N=4721	0.044 (0.054) [0.411] N=4761	-0.013 (0.043) [0.763] N=4761	-0.014 (0.038) [0.705] N=4485	0.023 (0.035) [0.509] N=4485
Panel C: Inverse-Variance Weighted-Average						
Anticipatory Action	-0.101** (0.044) [0.021]	0.179*** (0.052) [0.001]	0.005 (0.043) [0.901]	0.007 (0.039) [0.865]	-0.004 (0.032) [0.897]	0.020 (0.031) [0.516]
Post-flood mean:NEP	0.051	-0.079	0.030	-0.043	-0.011	-0.004
Post-flood mean:BGD	0.067	-0.103	-0.009	0.022	0.009	0.018
Post-flood mean:Pooled	0.060	-0.093	0.007	-0.006	-0.000	0.009
Strata X Round FE	Yes	Yes	Yes	Yes	Yes	Yes
N Clusters	428	428	438	438	438	438
N Observations	6933	6933	7126	7126	6893	6893

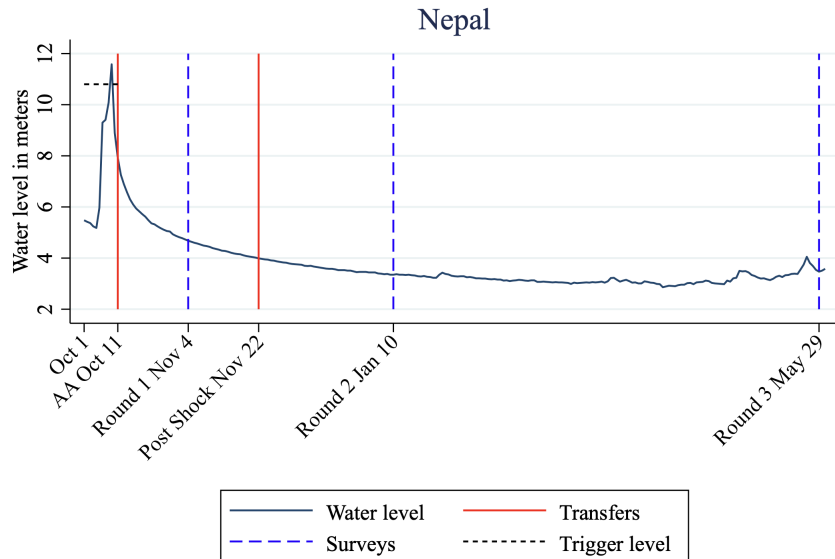
Notes: This table presents the impact of being in the FbAA group relative to the post-flood group on PHQ4 score and Cantril ladder across the three survey rounds. PHQ4 and Cantril ladder were standardized to create z-scores using mean and standard deviation within each round. Reduction in PHQ4 implies better mental health. Panel A shows results for Nepal only. Panel B shows results for Bangladesh only. Panel C shows an invest-variance weighted average of the results from Nepal and Bangladesh. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels.

Summary of findings

- Benefits of forecast-based cash transfers mainly come from realizing impacts as early as possible. Better food security and mental health in the immediate impact of the flood translate to positive overall impacts from FbF.
- Potential trade-off: Acting as quickly as possible has benefits relative to acting late, but may contribute to targeting errors. If forecasts have any error, we will sometimes end up giving transfers to people who are not the most affected or not affected at all. In what contexts is it better to just wait for the disaster to happen or only generate registries after the forecast triggers?

Thank you!

Appendix: livelihoods & flood heterogeneity in Nepal



Chisapani station measured water level.

Agriculture outcomes in subsequent season-Nepal

	(1) Planted	(2) # Crops	(3) Area	(4) Wheat Yield	(5) Costs	(6) Crop Value	(7) Sales Revenue
FbAA x Round 1							
FbAA x Round 2							
FbAA x Round 3	-0.05 (0.03)	0.04 (0.09)	0.03 (0.04)	0.09 (0.09)	0.02 (0.04)	30.10 (23.18)	15.08 (11.10)
Delayed group mean	0.76	1.91	0.33	1.93	0.34	166.48	47.33
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Muni x Round FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1983	1486	1983	739	1983	1983	1983

Notes: Standard errors are clustered at the village level. Baseline controls selected by post-double selection Lasso. Asterisks indicate statistical significance at the 1%***, 5%** , and 10%* levels.

On average, no impact of early transfers on agricultural outcomes

	(1) Planted	(2) # Crops	(3) Area	(4) Wheat Yield	(5) Costs	(6) Crop Value	(7) Sales Revenue
FbAA x R3	-0.07* (0.04)	0.05 (0.09)	0.00 (0.04)	0.05 (0.10)	-0.00 (0.04)	11.12 (23.24)	4.41 (10.83)
FbAA x Flood x R3	0.09 (0.06)	-0.00 (0.20)	0.11 (0.08)	0.12 (0.17)	0.13** (0.07)	85.26* (47.67)	45.52** (22.99)
Flood x R3	0.00 (0.04)	0.03 (0.14)	0.03 (0.05)	0.14 (0.12)	0.09* (0.05)	32.61 (28.12)	13.24 (14.23)
Delayed group mean	0.76	1.91	0.33	1.93	0.34	166.48	47.33
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Muni x Round FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1983	1486	1983	739	1983	1983	1983

Notes: Standard errors are clustered at the village level. Baseline controls selected by post-double selection Lasso. Asterisks indicate statistical significance at the 1%***, 5%** , and 10%* levels.

But, for the most flood affected households (as verified by remote sensing data), AA group had higher expenditure and earned more.

Impacts on livelihoods

Also find no impacts of transfer modality on:

- Wage income
- Earnings from livestock
- Business profits



Impact of AA on psychological wellbeing.



Impact of AA on psychological wellbeing. Inverted PhQ4 score.

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